HW 7

10.3 $E(r) = E_0 r_0 \dfrac{e^{ikr}}{r}$ $\qquad \nabla^2\psi = \dfrac{1}{r}\dfrac{\partial^2}{\partial r^2}(r\psi) + \dfrac{1}{r^2\sin\theta}\dfrac{\partial}{\partial\theta}\left(\sin\theta\dfrac{\partial\psi}{\partial\theta}\right) + \dfrac{1}{r^2\sin^2\theta}\dfrac{\partial^2\psi}{\partial\phi^2}$

$\nabla^2 E(r) + k^2 E(r) = 0 \qquad \nabla^2 E(r) = \dfrac{1}{r}\dfrac{\partial^2}{\partial r^2}(rE(r))$

$\hookrightarrow \dfrac{1}{r}\dfrac{\partial^2}{\partial r^2} rE(r) + k^2 E(r) = 0$

$u = E(r)r$

$\hookrightarrow \dfrac{\partial^2(E(r)r)}{\partial r^2} = -k^2 r\, E(r) \Rightarrow \dfrac{\partial^2 u}{\partial r^2} = -k^2 u$

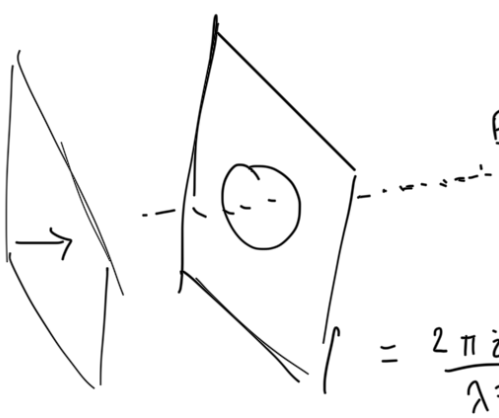
$\hookrightarrow u = Ae^{ikr} + Be^{-ikr} \Rightarrow E(r) = \dfrac{A}{r}e^{ikr} + \dfrac{B}{r}e^{-ikr}$

$B = 0$

$A = E_0 r_0$ $\Rightarrow \boxed{E(r) = E_0 r_0 \dfrac{e^{ikr}}{r}}$

10.8 a.) 10.1 using 10.27, 10.28

$\rho, x, y = 0 \qquad J_0(0) = 1$



$E(0,0,z)$

$$E(\rho,z) = \frac{2\pi i e^{ikz} e^{i\frac{k\rho^2}{2z}}}{\lambda z}\int_0^{D/2}\rho' d\rho' E(\rho',0)\, e^{\frac{ik\rho'^2}{2z}} J_0\left(\frac{k\rho\rho'}{z}\right)$$

$$= \frac{2\pi i e^{ikz}}{\lambda z}\int_0^{D/2}\rho' d\rho' E(\rho',0)\, e^{\frac{ik\rho'^2}{2z}}$$

$u = \rho'^2$

$du = 2\rho' d\rho'$

$$= \frac{\pi \cancel{i} e^{ikz}}{\lambda \cancel{z}}\frac{2\cancel{z}}{\cancel{i}k} E_0 e^{\frac{ik\rho'^2}{2z}}\Big|_0^{D/2}$$

$$= \frac{2\pi e^{ikz}}{\lambda k} E_0\left(e^{\frac{ik\,D^2/4}{2z}} - 1\right)$$

$I = E^* E \qquad |e^{i\theta} - 1|^2 = 4\sin^2(\theta/2)$

$\Rightarrow I = 4I_0 \sin^2\left(\dfrac{kD^2}{16z}\right)$

$$E(\rho, z) = -\frac{2\pi i e^{ikz} e^{i\frac{k\rho^2}{2z}}}{\lambda z} \int \rho' d\rho' \, E(\rho', 0) \, J_0\left(\frac{k\rho\rho'}{z}\right)$$

$$= -\frac{\pi i e^{ikz} e^{i\frac{k\rho^2}{2z}}}{\lambda z} E_0 \rho'^2 \Big|_0^{D/2} = -\frac{\pi i e^{ikz} e^{i\frac{k\rho^2}{2z}}}{\lambda z} E_0 \frac{D^2}{4}$$

$I = E^* E$

$\hookrightarrow I = I_0 \left(\frac{\pi D^2}{4\lambda z}\right)^2$

b.) $\sqrt{1+\alpha} \approx 1 + \frac{\alpha}{2}$ $\quad \cos\beta \approx 1 + \beta^2/2$

$$E(0,0,z) E^*(0,0,z) = 2|E_0|^2 \left[1 - \cos\left(k\sqrt{(D/2)^2 + z^2} - kz\right)\right]$$

$$z\left(\frac{D^2}{z^2 4} + 1\right)^{1/2}$$

$\hookrightarrow z\left(1 + \frac{D^2}{8z^2}\right)$

$\cos\left(zk + \frac{kD^2}{8z} - kz\right)$

$\hookrightarrow 1 - \cos\left(\frac{kD^2}{8z}\right)$

$\hookrightarrow I = 4I_0 \sin^2\left(\frac{kD^2}{16z}\right)$

$1 - 1 - \frac{k^2 D^4}{128 z^2} \Rightarrow I = -I_0 \frac{k^2 D^4}{64 z^2}$ $\quad k = \frac{2\pi}{\lambda}$

$\hookrightarrow I = I_0 \frac{\pi^2 D^4}{16 \lambda z^2}$